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COURSE NAME: ANALOG AND DIGITAL COMMUNICATION (ECA0802)

EX NO: 3a

GENERATION OF PHASE MODULATED (PM) SIGNALS AND
CALCULATION OF PHASE DEVIATION USING MATLAB
SIMULATION

Objective

- i) To generate a Phase Modulated (PM) signal in MATLAB using specified carrier and message signal parameters.
- ii) To observe and analyze the time-domain characteristics of the message signal, carrier signal, and phase modulated signal through MATLAB simulation.
- iii) To calculate the phase deviation of the PM signal using the phase sensitivity constant and message signal amplitude.
- iv) To determine the modulation index of the PM signal and examine its relationship with phase deviation.
- v) To investigate the effect of varying message signal amplitude on the phase deviation and modulation characteristics of the PM signal.
- vi) To compare PM waveforms obtained for different phase deviation values and analyze their modulation behavior using MATLAB.

PROGRAM (MATLAB CODE)

```
clc;
```

```
clear;
```

```
close all;
```

```
% Parameters
```

```

Am = 1;
fm = 100;
Ac = 5;
fc = 1000;
fs = 20*fc;
kp = pi/2;
% Message Amplitude (V)
% Message Frequency (Hz)
% Carrier Amplitude (V)
% Carrier Frequency (Hz)
% Sampling Frequency (Hz)
% Phase Sensitivity (rad/V)
% Time Vector
t = 0:1/fs:4/fm;
% Message Signal
m = Am*sin(2*pi*fm*t);
% Carrier Signal
c = Ac*cos(2*pi*fc*t);
% Phase Modulated Signal
pm = Ac*cos(2*pi*fc*t + kp*m);
% Phase Deviation
phase_dev = kp*Am;
% PM Modulation Index
mp = phase_dev;
% Display Results
fprintf('Message Amplitude = %.2f V\n',Am);
fprintf('Message Frequency = %.0f Hz\n',fm);
fprintf('Carrier Amplitude = %.2f V\n',Ac);
fprintf('Carrier Frequency = %.0f Hz\n',fc);

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fprintf('Phase Sensitivity = %.2f rad/V\n',kp);
fprintf('Phase Deviation = %.2f radians\n',phase_dev);
fprintf('PM Modulation Index = %.2f\n',mp);

% Plotting

figure('Name','Phase Modulation and Phase Deviation',...
'NumberTitle','off');

subplot(2,2,1);

plot(t,m,'LineWidth',1.5);

title('Message Signal');

xlabel('Time (s)');

ylabel('Amplitude');

grid on;

subplot(2,2,2);

plot(t,c,'LineWidth',1.5);

title('Carrier Signal');

xlabel('Time (s)');

ylabel('Amplitude');

grid on;

subplot(2,2,3);

plot(t,pm,'LineWidth',1.5);

title('Phase Modulated Signal');

xlabel('Time (s)');

ylabel('Amplitude');

grid on;

subplot(2,2,4);

bar(phase_dev);

title('Phase Deviation');

ylabel('Radians');

set(gca,'XTickLabel',{'PM Signal'});

```

grid on;

OUTPUT:

